

8-2-1: Examining VXLAN

At the end of this episode, I will be able to:

1. Identify and explain the importance of VXLAN technologies, given a scenario.

Learner Objective: *Identify and explain the importance of VXLAN technologies, given a scenario*

Description: In this episode, the learner will examine VXLAN. We will explore VXLAN technologies, and a comparison to traditional VLANs.

- What is Virtual Extensible LAN (VXLAN)?
 - A network virtualization technology designed to provide scalable, isolated networks in cloud computing environments.
 - Provides significantly more segmentation options than the 4096 VLAN IDs available in traditional VLAN.
 - What are the comparisons to traditional VLANs?
 - o Scalability
 - o Cross Data Center Connectivity
 - o Network Segmentation
 - Taking each one of these, how do they benefit networks?
 - o Scalability
 - ♠ VLANs are limited to 4096 unique IDs, whereas VXLAN supports up to 16 million VNIs, offering greater scalability.
 - o Cross Data Center Connectivity
 - ♠ VXLAN can stretch Layer 2 networks across geographically dispersed data centers, unlike VLANs, which are typically confined to a single physical location.
 - o Network Segmentation
 - ♠ Provides enhanced network segmentation capabilities, essential in multi-tenant or cloud environments where VLAN's capabilities might be insufficient.
 - What are some of the limitations overcome by VXLAN?
 - o Address Space Limitation
 - o Layer 2 Extension
 - o Multi-Tenancy Support
 - Taking each one of these, how do they help overcome some of the limitations of VLANs?
 - o Address Space Limitation
 - ♠ Overcomes the 4096 VLAN ID limitation, addressing the needs of larger and more dynamic environments.
 - o Layer 2 Extension
 - ♠ Enables Layer 2 network extension over Layer 3 infrastructure, facilitating more flexible and efficient network designs.
 - o Multi-Tenancy Support
 - ♠ Better supports multi-tenancy environments, which is crucial for modern data centers and cloud services, where traditional VLANs may fall short.
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- Additional Resources:
 - o If applicable

